

Course Code: CAP 680	Course Instructor: Mr. Kumar Vishal
Course Title: Programming in Java Laboratory	
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Page Number: 01 Total Pages: 08	

Q1: Suppose there is the string "Lovely Professional University" define some methods which

will be use to perform

a. finds the total no. of characters

b. finds the position of string University

c. extracts the string Lovely

d. replaces Lovely with lovely (first letter 'l' in small case)

e. changes all characters in the string in Upper case.

Answer: -

```
class StringExample
```

```
{
```

```
String str1 = "Lovely Professional University";
```

```
public void display()
```

```
{  
    int length=str1.length();           //Counts each  
    character including space  
        System.out.println("Total number of  
characters present in the string : " + length);  
    String str2 = "University";  
    int index = str1.indexOf(str2);  
        System.out.println("The position of string  
University is at : " + index +" index");  
    String res = "";  
    res = str1.substring(0,6);  
        System.out.println("The extracted string is : "  
+ res);  
    String rep="";  
    rep = str1.replaceAll("Lovely","lovely");  
        System.out.println("After replacing Lovely  
with lovely :"+rep );  
    String upCase="";  
    upCase = str1.toUpperCase();
```

```
        System.out.println("changing all characters  
in Upper case : "+ upCase);  
    }  
}  
  
public class Main  
{  
  
    public static void main(String[] args) {  
        StringExample ob = new StringExample();  
        ob.display();  
    }  
}
```

```
Total number of characters present in the string : 30  
The position of string University is at : 20 index  
The extracted string is : Lovely  
After replacing Lovely with lovely :lovely Professional University  
changing all characters in Upper case : LOVELY PROFESSIONAL UNIVERSITY  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

Q2: Create a class named 'Time' with data members (hours, mins, secs). Define methods in this

class named 'hourConvertor', 'minConvertor' and 'secConvertor'. Calculate total hours, mins and secs for any integer value (in days).

For eg:

Input : 2 (days)

output: total_hours:48

total_mins:2880

total_secs:172800

Answer: -

```
import java.util.Scanner;  
class time  
{  
    public int hours, mins, secs;  
    public int hourConvertor(int day)  
    {
```

```
hours = day*24;
    return hours;
}

public int minConvertor(int day)
{
    mins = day*24*60;
    return mins;
}

public int secConvertor(int day)
{
    secs = day*24*60*60;
    return secs;
}
}

public class Main
{
    public static void main(String[] args) {
        time t = new time();
        Scanner in = new Scanner(System.in);
```

```
int day;  
System.out.println("Enter number of days :");  
day = in.nextInt();  
System.out.println("hours:"+t.hourConvertor(da  
y));  
System.out.println("minutes:"+t.minConvertor(d  
ay));  
System.out.println("seconds:"+t.secConvertor(da  
y));  
    }  
}
```

```
Enter number of days :2  
hours : 48  
minutes : 2880  
seconds : 172800
```

```
...Program finished with exit code 0  
Press ENTER to exit console.□
```

Q3: Write a Java program that checks whether a given string is a palindrome or not.

Ex: MADAM is a palindrome?

Answer: -

```
import java.util.*;

class Main
{
    public static void main(String args[])
    {
        String original, reverse = "";
        Scanner in = new Scanner(System.in);
        System.out.println("Enter a string to check
palindrome:");
        original = in.nextLine();
        int length = original.length();
        for ( int i = length - 1; i >= 0; i-- )
            reverse = reverse + original.charAt(i);
        if (original.equals(reverse))
```

```
System.out.println("Entered string is a  
palindrome.");  
    else  
System.out.println("Entered string isn't a  
palindrome.");  
}  
}
```

```
Enter a string to check palindrome:  
MADAM  
Entered string is a palindrome.  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```